

not correctly account for the dependence between  $s_i^A$  and  $s_{ir}^E$ , which we call *signal dependence neglect*. For example, if humans act as-if  $s_i^A$  and  $s_{ir}^E$  are independent conditional on  $\omega_i$  even if they are not, then their posterior beliefs can be written as

$$\log \frac{p_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{p_r(\omega_i = 0 | s_i^A, s_{ir}^E)} = b_r \log \frac{\pi_r(s_i^A | \omega_i = 1)}{\pi_r(s_i^A | \omega_i = 0)} + d_r \log \frac{\pi_r(\omega_i = 1 | s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^E)}, \quad (6)$$

where  $b_r$  and  $d_r$  are allowed to differ from 1 as above. In the case when the signals are jointly multivariate normal and  $b_r = d_r = 1$ , signal dependence neglect yields correlation neglect as defined in (Enke and Zimmermann, 2019).<sup>26</sup> More generally, we will consider models that vary the conditioning set in the first term on the right-hand side and the dimension of  $s_i^A$  in the first term on the right-hand side. The specific examples are motivated and discussed further in section 5.3 below.

We intend the functional form above as descriptions of humans' updating rules and will remain agnostic about underlying mechanisms and micro-foundations. In particular, we remain silent on whether our participants are Bayesians that are utilizing the incorrect joint distribution of  $(\omega_i, s_i^A, s_{ir}^E)$  when updating their beliefs or if they are non-Bayesians. The former type of model, known as a quasi-Bayesian model,<sup>27</sup> can generate automation bias or neglect as well as correlation biases.<sup>28</sup> An implicit assumption in our model, and likely other micro-foundations for the functional forms above as well, is that the signal acquired by the human is invariant to the provision of AI assistance. Whether additional training, or experience with the AI can correct deviations from the benchmark model is therefore something that we leave for future work.

Nonetheless, the models above will prove useful for our purposes. From a theoretical perspective, they will help outline the types of deviations that potentially decrease decision quality. From an empirical perspective, they help understand the drivers of the treatment effects documented earlier and turn out to be a good approximation to the data from the experiment.

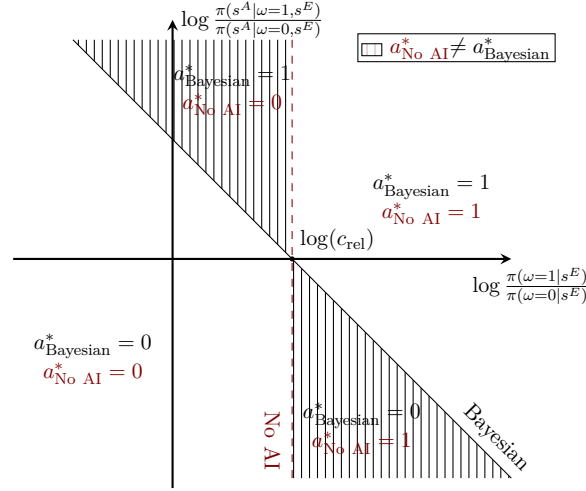
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<sup>26</sup>If  $s_i^A$  and  $s_{ir}^E$  are unidimensional with  $(s_i^A, s_{ir}^E) \sim N(0, \Sigma_r)$ , then the covariance matrix  $\Sigma_r$  is a sufficient statistic for the posterior probability that  $\omega_i = 1$  given the signals if  $\omega_i = 1 \{s_i^A + s_{ir}^E \geq \varepsilon_i\}$  and  $\varepsilon_i$  is independent of  $(s_i^A, s_{ir}^E)$ .

<sup>27</sup>See Rabin (2013) for a definition and Barberis et al. (1998); Rabin (2002); Rabin and Vayanos (2010) for examples.

<sup>28</sup>To see this, assume that  $p_r(s_i^A | \omega_i, s_{ir}^E) = \pi(s_i^A | \omega_i, s_{ir}^E)^b$  and  $p_r(s_{ir}^E, \omega_i) = \pi_r(s_{ir}^E, \omega_i)$  to generate the functional form in equation (5) for any  $b_r$  as long as  $d_r = 1$ . The derivation of equation (6) is similar. In contrast to automation bias/neglect and correlation biases, own-information bias/neglect cannot be derived in a quasi-Bayesian model because we assume that  $p_r(\omega_i | s_{ir}^E) = \pi_r(\omega_i | s_{ir}^E)$ .

Figure 4: Comparing decisions with and without AI assistance – Bayesian with correct beliefs



Note: The figure shows the decision criterion of a Bayesian with and without AI assistance and where their decisions agree and disagree. Shaded regions show the regions in which AI improves or worsens decision making.

## 5.2 Implications for Human-AI Collaboration

We now show that the types of deviations described above have implications for when AI assistance unambiguously improves human performance. The results will also illustrate the utility of the simple functional forms in equations (5) and (6). This subsection drops the  $i$  and  $r$  indices for simplicity of notation.

It is useful to start by considering the decisions with and without AI assistance for a Bayesian decision maker. Figure 4 illustrates the realizations of  $s^A$  for which the optimal decision with AI assistance differs from the the decision without AI assistance for a fixed  $c_{rel}$ . The horizontal and vertical axes respectively represent  $\log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$  and  $\log \frac{\pi(s^A|\omega=1,s^E)}{\pi(s^A|\omega=0,s^E)}$ . As shown by the vertical dashed line, the decision-maker would take the action 1 if and only if  $\log \frac{\pi(\omega=1|s^E)}{\pi(\omega=0|s^E)}$  exceeds  $\log c_{rel}$ . The solid line represents the analogous boundary for a Bayesian who has access to AI assistance. Observe that the decisions a Bayesian makes as a function of the signals  $s^A$  and  $s^E$  cannot be improved without additional information. Thus, a Bayesian with correct beliefs and access to both signals improves upon the no AI action in the vertically shaded region.

Now consider humans that may deviate from this benchmark model. A human who takes a given action without AI assistance  $a^*_{No AI} = a^*(s^E; p_r)$ , but a different action with AI assistance (so that  $a^*_{No AI} \neq a^*_{AI}$ ) makes a worse decision if  $a^*_{AI} = a^*(s^A, s^E; p_r)$  disagrees with

the Bayesian's decision  $a_{\text{Bayesian}}^* = a^*(s^A, s^E; \pi_r)$  with AI assistance. This follows because, in the binary action setup, only one of the decisions can agree with the Bayesian decision. In all other cases, the human's decision is weakly improved for the signal realization  $(s^A, s^E)$ . In other words, a human whose decision changes upon receiving the AI signal  $s^A$  is better off with AI assistance only if the change agrees with the Bayesian decision. The human is unambiguously better off if this property holds for all signals.

Our first result states that a human who deviates from the benchmark model because they only exhibits automation neglect is unambiguously better off with AI assistance.

**Proposition 1.** *Suppose that the human's posterior is described by equation (5).*

(i) *If the human only exhibits automation neglect ( $b < 1, d = 1$ ), then for all pairs of signal realizations  $(s^A, s^E)$ , and any  $c_{rel}$ , the human attains weakly higher expected payoff  $V(s)$  with AI assistance.*

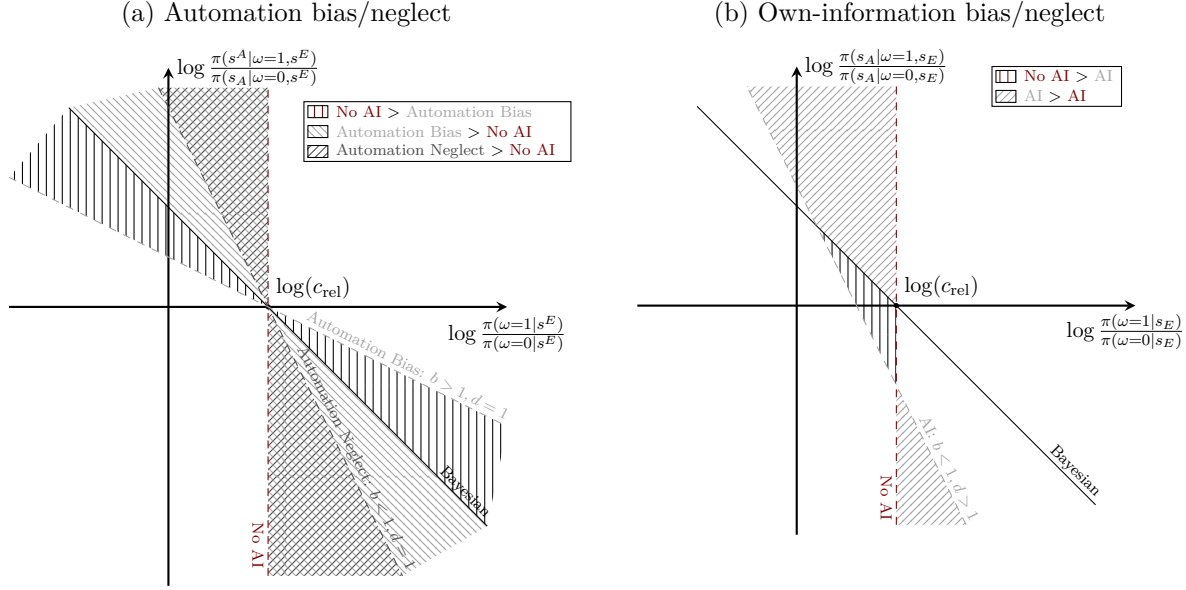
(ii) *In all other cases ( $b > 1$  or  $d \neq 1$ ), for any  $c_{rel}$ , there exist log-likelihood ratios  $\log \frac{\pi(s^A | \omega=1, s^E)}{\pi(s^A | \omega=0, s^E)}$  and  $\log \frac{\pi(\omega=1 | s^E)}{\pi(\omega=0 | s^E)}$  such that the human attains lower expected payoff  $V(s)$  with AI assistance.*

See appendix A for the proof.

Figure 5a illustrates the case without own-information bias or neglect by setting  $d = 1$  but allows for either automation bias or neglect by setting  $b \neq 1$ . The two dashed lines represent cutoffs analogous to those in figure 4 for humans with automation bias and automation neglect. Although a human that only exhibits automation neglect under-responds to the AI information, their beliefs move towards a Bayesian decision-makers but does not overshoot it. Whenever her decision changes, it agrees with the Bayesian's. In contrast, if the human exhibits automation bias, they errs for moderately informative AI signals with intermediate values of  $\log \frac{\pi(s^A | \omega=1, s^E)}{\pi(s^A | \omega=0, s^E)}$  because they over-reacts. At high enough values of this log-likelihood ratio, both the Bayesian and the human exhibiting automation bias would take the same action.

In the case when there is either own-information bias or neglect ( $d \neq 1$ ), it is always possible for the payoff with AI to be lower than the payoff without AI assistance. In our model, this occurs because providing the AI signal generates a bias in how the expert uses their own information. Figure 5b illustrates the result (for  $d > 1$  and  $b < 1$ ). Relative to the analogous line in figure 5a, the intercept of the dashed line is not equal to  $\log c_{rel}$ . The vertically striped triangle depicts combinations of the log-likelihood ratios above in which the biased expert achieves a higher payoff without AI. In these cases, the expert takes the correct action without AI assistance but incorrectly uses their own information with AI assistance, resulting in an

Figure 5: Comparison with Bayesian decisions



Note: This figure shows the where the decisions of an expert as a function of the signals disagree with the Bayesian in cases with and without AI assistance. Panel (a) shows automation bias and neglect (absent own-information bias and neglect). Panel (b) shows a decision maker who exhibits both automation neglect and own-information bias.

error. Hence, in the presence of own-information bias or neglect, AI assistance can result in worse decision for some signals.

We next consider a decision-maker with a different type of bias, namely, one in which the decision-maker exhibits signal dependence neglect. Perhaps not surprisingly, our next result shows that this type of bias on its own can result in worse decisions with AI assistance:

**Proposition 2.** *Suppose that the human exhibits signal dependence neglect so that the posterior belief is described by equation (6). For any value of  $b > 0$ ,  $d > 0$ , and  $c_{rel} > 0$ , there exist log-likelihood ratios  $\log \frac{\pi(s^A|\omega=1,s^E)}{\pi(s^A|\omega=0,s^E)}$  and  $\log \frac{\pi(s^A|\omega=1)}{\pi(s^A|\omega=0)}$  such that the human attains lower expected payoff  $V(s)$  with AI assistance.*

Compared to the models of only automation/own-information bias/neglect, signal dependence neglect adds another dimension of potential mistakes to those illustrated in the figures above. The result bears resemblance to those in [Enke and Zimmermann \(2019\)](#), which showed that in a multivariate normal model with positively correlated signals, correlation neglect results in over-reaction to signals and verified this hypothesis in lab experiments. Proposition 2 differs in that it allows for general signal distributions and for signal dependence neglect to co-exist with automation and own-information bias/neglect. This extension is essential for a naturalistic environment like ours because the experimenter does not have full control over